LESEDI RENEWABLE ENERGY RESEARCH GROUP

Enterprise Network Design & Implementation Report

Segmented VLAN Architecture with Router-on-a-Stick Inter-VLAN Routing

Prepared for: Lesedi Renewable Energy Research Group, Mahikeng

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1.0	Aug 2026	Milestone 1	L.I Phiri	Initial requirements analysis, network design, and VLSM addressing plan
2.0	Sep 2026	Milestone 2	L.I Phiri	Full implementation in Cisco Packet Tracer; configuration, testing, and troubleshooting
3.0	Oct 2026	Milestone 3	L.I Phiri	Consolidated final report, verification evidence, and recommendations

Distribution

Role	Name / Organisation
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Referenced Documents

- Client requirements brief: CMPG325 CLI-109 assignment specification
- Physical and logical topology diagrams: project repository, /design
- VLSM IP addressing plan: project repository, /ip-addressing
- Cisco Packet Tracer project file: CMPG325-CLI-109-Lesedi-Network.pkt
- Project evidence repository: GitHub: CMPG325-CLI-109-Lesedi-Network

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Executive Summary

Lesedi Renewable Energy Research Group engaged this project to design, implement, and verify a segmented local area network for its Mahikeng research facility. The organisation required logical separation between research, administrative, and server traffic; centralised inter-VLAN routing through a single router interface; restricted, limited-purpose wireless access for after-hours contractors; and full compliance with the heritage-listed building's prohibition on new external cabling.

The delivered solution uses a five-VLAN architecture addressed from the client's assigned 172.30.74.0/23 block via Variable Length Subnet Masking (VLSM), routed entirely through a single Router-on-a-Stick configuration on Router-Lesedi. A dedicated wireless access point serves the restricted contractor VLAN over existing internal cabling, with an extended access control list enforcing selective denial of internal resources while preserving outbound connectivity.

The design was implemented and verified in Cisco Packet Tracer across three project milestones: requirements analysis and design (Milestone 1), implementation and testing (Milestone 2), and final consolidation and verification (Milestone 3, this report). Every client requirement (R1-R7) was traced from the original brief through to a specific, evidenced test result including direct proof of inter-VLAN routing via TTL analysis, and before/after verification that the contractor VLAN's access control list correctly restricts internal access while preserving Internet reachability.

This report consolidates the full project record: requirements, design rationale, implementation detail, verification evidence, issues encountered and resolved, and recommendations for future enhancement. Supporting evidence (configuration exports, screenshots, and the working Packet Tracer file) is maintained in the accompanying GitHub repository and video demonstration.

1. Introduction

1.1 Background

Lesedi Renewable Energy Research Group is a research organisation operating from a heritage-listed building in Mahikeng. The organisation's network previously had no formal segmentation between its research, administrative, and server functions, and lacked any mechanism for providing controlled, limited network access to non-staff personnel such as after-hours cleaning and security contractors.

1.2 Purpose and Scope of this Document

This report documents the complete lifecycle of the network design and implementation project commissioned under module CMPG325, from initial requirements capture through to final verified deployment in a simulated environment. It is intended to serve as the authoritative technical record of the project, suitable for technical review, academic assessment, and as a reference should the network be extended in future.

The scope of this report covers: client requirements and their traceability; the logical and physical network design, including IP addressing; the configuration applied to every network device; the verification testing performed and its results; issues encountered during implementation and their resolution; and recommendations for future work.

1.3 Project Methodology

The project was delivered across three milestones, each building directly on the last without deviation from the approved design:

- Milestone 1: Requirements analysis, network design, and IP addressing plan.
- Milestone 2: Full implementation in Cisco Packet Tracer, configuration of all switching and routing infrastructure, and verification testing.
- Milestone 3: Final consolidation, independent re-verification of all evidence, and production of this report together with a recorded technical demonstration.

2. Client Requirements

Each requirement below is traced directly to the client brief or an explicit client change request; no requirement was introduced beyond what was specified, and none was descoped without justification (see Section 2.2).

2.1 Requirements Traceability

Ref	Requirement	Source
R1	All internal addressing must use the assigned 172.30.74.0/23 block.	Fixed allocation, assignment brief
R2	Traffic must be logically segmented by function (research, administration, servers/lab equipment).	Inferred from organisational structure of a research group with distinct staff and lab/server functions
R3	Inter-VLAN communication must be routed through a single router using one 802.1Q sub-interface per VLAN (Router-on-a-Stick).	Assigned technical challenge, mandatory
R4	After-hours cleaning and security contractors require wireless network access, limited rather than full.	Explicit client change request
R5	No new cabling may be run through the building's external walls.	Heritage-listed building constraint
R6	The design must be fully implementable and testable in Cisco Packet Tracer, producing a working project file.	Assignment deliverable requirement
R7	End-to-end connectivity must be demonstrable for intra-VLAN, inter-VLAN, and correctly restricted contractor traffic.	Assignment testing requirement, combined with R4

2.2 Scope Boundaries

The following were considered and deliberately excluded from scope, as they were not requested in the brief and would have introduced unnecessary complexity relative to the client's stated needs:

- Additional VLANs or additional physical sites
- Full network access control (802.1X) for guest/contractor authentication
- Router or switch redundancy (HSRP/VRRP) or other high-availability mechanisms
- Dynamic routing protocols, a single router with static sub-interface routing is sufficient for a single-site, five-VLAN network
- A wired guest network, the contractor requirement was specifically for wireless access

3. Network Design

3.1 Physical Topology

The physical design centres on a single router (Router-Lesedi) connected by one trunk link to a Core Switch, which in turn aggregates two access switches, two directly attached servers, and (indirectly, via an access switch) a wireless access point.

Device	Role	Key Configuration Notes
Router-Lesedi	Inter-VLAN routing (Router-on-a-Stick)	Single Gi0/0 trunk to Core Switch; five 802.1Q sub-interfaces, one per VLAN
Core-Switch	Trunk aggregation	Trunk to router; trunk to each access switch; access ports (VLAN 30) to both servers
SwitchA-Research	Access layer, Research wing	VLAN 10 access ports; VLAN 40 access port (AP uplink)
SwitchB-Admin	Access layer, Admin office	VLAN 20 access ports
AP-GUEST	Restricted wireless access	Connected via existing internal cabling on SwitchA-Research, no new external cable run

3.1.1 Heritage Building Constraint (R5)

As the facility is heritage-listed, no new cabling may penetrate an external wall. This was satisfied by mounting the contractor-facing wireless access point indoors, fed from an existing internal access-switch port, so wireless coverage extends network access into the building without any new external cable run. All other new connectivity (the router-to-switch trunk and inter-switch links) likewise uses existing internal risers and conduits.

3.2 Logical Design: VLAN Strategy

Five VLANs separate traffic by function, with all inter-VLAN routing centralised on Router-Lesedi via Router-on-a-Stick, directly satisfying the assigned technical challenge (R3) while remaining proportionate to a single-site, moderate-traffic network (R6).

VLAN	Name	Purpose
10	RESEARCH	Researcher PCs and lab/data-collection workstations
20	ADMIN	Administration and management staff
30	SERVERS	File server, research application server, lab equipment
40	GUEST-WIFI	Restricted after-hours contractor wireless access
99	MGMT	Switch and access-point management (no end-user hosts)

3.3 IP Addressing Plan: VLSM

The client's assigned 172.30.74.0/23 block (512 addresses, 172.30.74.0 - 172.30.75.255) was subdivided using Variable Length Subnet Masking, sizing each subnet to its realistic host population

rather than cutting the block into equal, wasteful shares. Roughly half the block (172.30.75.16 onward) remains unallocated and reserved for future growth.

VLAN	Subnet	Gateway	Usable Hosts	Sizing Rationale
10	172.30.74.0/25	172.30.74.1	126	Largest functional group, research and lab PCs
20	172.30.74.128/26	172.30.74.129	62	Administrative staff, with headroom
30	172.30.74.192/27	172.30.74.193	30	Small, well-defined server population
40	172.30.74.224/27	172.30.74.225	30	Small, restricted contractor population
99	172.30.75.0/28	172.30.75.1	14	Infrastructure only, no end-user hosts

3.4 Design Justification

3.4.1 Why Router-on-a-Stick

A single router interface, logically divided via 802.1Q sub-interfaces, was selected over a multilayer switch because: it directly satisfies the assigned technical challenge (R3); the network's scale, a single site, five VLANs, moderate traffic volume, does not justify the additional hardware cost or complexity of a multilayer switch; and it keeps the design fully verifiable within Cisco Packet Tracer (R6).

3.4.2 Enforcing Restricted Contractor Access (R4)

VLAN 40 hosts obtain the router's Fa0/0.40 sub-interface as their default gateway identically to any other VLAN, but an extended access control list on that sub-interface permits outbound traffic (e.g. to the Internet/ISP) while explicitly denying traffic destined for VLANs 10, 20, 30, and 99. This satisfies the client's explicit requirement for "limited," rather than absent, wireless access, full implementation and verification detail is provided in Sections 4.3 and 5.4.

4. Implementation

All configuration below was applied and verified directly in Cisco Packet Tracer, with no deviation from the design approved in Section 3.

4.1 Switching Infrastructure

4.1.1 Core-Switch

- VLANs 10, 20, 30, 40, and 99 created.
- 802.1Q trunk configured on the uplink to Router-Lesedi, carrying all five VLANs.
- 802.1Q trunks configured to SwitchA-Research and SwitchB-Admin.
- Access ports assigned to VLAN 30 for Server-File and Server-App.

4.1.2 SwitchA-Research

- VLAN 10 access ports for PC-Research1 and PC-Research2.
- VLAN 40 access port for the AP-GUEST uplink.
- Trunk link to Core-Switch.

4.1.3 SwitchB-Admin

- VLAN 20 access port for PC-Admin1, confirmed via show vlan brief.
- Trunk link to Core-Switch.

4.2 Router-on-a-Stick Configuration (Router-Lesedi)

Five sub-interfaces were created on FastEthernet0/0, each carrying 802.1Q encapsulation for its VLAN and configured as that VLAN's default gateway:

```
interface FastEthernet0/0.10
encapsulation dot1Q 10
ip address 172.30.74.1 255.255.255.128

interface FastEthernet0/0.20
encapsulation dot1Q 20
ip address 172.30.74.129 255.255.255.192

interface FastEthernet0/0.30
encapsulation dot1Q 30
ip address 172.30.74.193 255.255.255.224

interface FastEthernet0/0.40
encapsulation dot1Q 40
ip address 172.30.74.225 255.255.255.224

interface FastEthernet0/0.99
encapsulation dot1Q 99
ip address 172.30.75.1 255.255.255.240

interface FastEthernet0/0
no shutdown
```

A sixth, independent interface (FastEthernet1/0, 203.0.113.1/30) connects to a Server-Internet node, representing the ISP boundary used to verify that VLAN 40 retains permitted outbound connectivity after restriction (Section 5.4).

4.3 Restricted Wireless Access : VLAN 40 (R4)

4.3.1 Wireless Access Point

- AP-GUEST configured with SSID "Lesedi-Guest"; Port 1 (Radio) enabled.
- Authentication left open at the wireless layer; the substantive access control is enforced at Layer 3 (Section 4.3.2), which is sufficient and appropriate for this network's scale.
- Laptop-Contractor fitted with a wireless network interface and connected to the "Lesedi-Guest" SSID, statically addressed at 172.30.74.226/27, gateway 172.30.74.225.

4.3.2 Access Control List 140

```
access-list 140 deny ip 172.30.74.224 0.0.0.31
172.30.74.0 0.0.0.127
access-list 140 deny ip 172.30.74.224 0.0.0.31
172.30.74.128 0.0.0.63
access-list 140 deny ip 172.30.74.224 0.0.0.31
172.30.74.192 0.0.0.31
access-list 140 deny ip 172.30.74.224 0.0.0.31
172.30.75.0 0.0.0.15
access-list 140 permit ip any any
```

ACL Statement	**Effect**
deny ... 172.30.74.0 0.0.0.127	Denies VLAN 40 access to VLAN 10 (Research, /25)
deny ... 172.30.74.128 0.0.0.63	Denies VLAN 40 access to VLAN 20 (Admin, /26)
deny ... 172.30.74.192 0.0.0.31	Denies VLAN 40 access to VLAN 30 (Servers, /27)
deny ... 172.30.75.0 0.0.0.15	Denies VLAN 40 access to VLAN 99 (Management, /28)
permit ip any any	Preserves all other connectivity, including outbound Internet access

The ACL is applied inbound on Fa0/0.40, filtering traffic as it enters the router from VLAN 40 devices. This direction was deliberately chosen: an inbound ACL on the source VLAN's own gateway interface filters the traffic at the earliest possible point, before it can be routed anywhere else in the network.

5. Verification & Testing

5.1 Test Methodology

Every VLAN and the access restriction policy were verified using ICMP (ping) testing from each end device, supported by router and switch show-command output. Testing proceeded in four stages: intra-VLAN connectivity, inter-VLAN routing, restricted-VLAN policy (before and after the ACL), and ACL match-counter confirmation. Full screenshot evidence for every test below is maintained in the project's GitHub repository (/screenshots) and demonstrated live in the accompanying video.

5.2 Intra-VLAN Connectivity

Device	VLAN	Target (Gateway)	Result
PC-Admin1	20	172.30.74.129	4/4 replies, 0% loss, TTL=255
PC-Research1 / PC-Research2	10	172.30.74.1	4/4 replies, 0% loss, TTL=255
Server-File / Server-App	30	172.30.74.193	4/4 replies, 0% loss, TTL=255
Laptop-Contractor (wireless)	40	172.30.74.225	4/4 replies, 0% loss

TTL=255 on each result reflects Cisco IOS's default TTL, confirming the reply originated directly from the router's own gateway interface for that VLAN, basic Layer 2/3 reachability is confirmed for every VLAN.

5.3 Inter-VLAN Routing Verification

PC-Admin1 (VLAN 20) was pinged against Server-File (VLAN 30, 172.30.74.194): 4/4 replies, 0% loss, with TTL reduced to 127.

5.4 Restricted VLAN 40: Before and After ACL 140

5.4.1 Before Restriction

Prior to applying ACL 140, Laptop-Contractor (VLAN 40) successfully reached Server-File (VLAN 30): 4/4 replies, TTL=127, confirming the network was fully open prior to the restriction being configured, and establishing a clean baseline.

5.4.2 After Restriction

With ACL 140 applied to Fa0/0.40, the identical test was repeated against Server-File, PC-Research1 (VLAN 10), and PC-Admin1 (VLAN 20). In every case the result was "Destination host unreachable," returned explicitly by the router's own VLAN 40 gateway address (172.30.74.225), confirming the router is actively denying the traffic by policy, rather than the traffic being silently lost.

5.4.3 Preserved Internet Access

Laptop-Contractor was then pinged against the Server-Internet node (203.0.113.2), confirming continued outbound connectivity: 3/4 replies succeeded with TTL=127 (the single initial timeout is the expected one-off ARP resolution delay on first contact, not a fault). This demonstrates the restriction is selective rather than total, directly satisfying the client's requirement for "limited," not absent, contractor access (R4).

5.5 Access Control List Verification

show access-lists 140, executed after the above tests, returned match counters against every statement: four matches on each deny line (corresponding to the four blocked test pings per denied VLAN) and four matches on the permit line (corresponding to the permitted Internet-bound test traffic). This confirms the ACL is actively filtering live traffic, consistent with the behaviour observed in Sections 5.4.2 and 5.4.3, rather than being present but inactive.

5.6 Summary of Verification Against Requirements

Requirement	**Verification Evidence**
R1: Assigned addressing block	All device and gateway addresses verified within 172.30.74.0/23 (Section 3.3)
R2: Logical segmentation	Five VLANs confirmed via show vlan brief across all three switches (Section 4.1)
R3: Router-on-a-Stick	Five sub-interfaces confirmed up/up with correct addressing (Section 4.2); inter-VLAN routing proven via TTL analysis (Section 5.3)
R4: Limited contractor access	ACL 140 verified denying internal VLANs while preserving Internet access (Section 5.4)
R5: No external-wall cabling	AP-GUEST connected via existing internal cabling (Section 3.1.1)
R6: Fully testable in Packet Tracer	Entire network built, configured, and tested within Cisco Packet Tracer
R7: End-to-end connectivity demonstrated	Intra-VLAN, inter-VLAN, and restricted-VLAN connectivity all evidenced (Sections 5.2-5.4)

6. Issues Identified and Resolved

The following issues were encountered during implementation and are documented here as part of the engineering record, consistent with standard practice for a professional deployment.

Issue	Root Cause	Resolution
Repeated "Invalid input" errors on router sub-interface IP address commands	Subnet mask octets were truncated during command entry (e.g. 255.255.192 instead of 255.255.255.192)	Commands re-entered deliberately in segments, with explicit verification of all four octets before execution
VLAN 10 sub-interface absent from show ip interface brief after a CLI session timeout	Configuration had not yet been committed to startup-config when the session timed out	Sub-interface recreated and immediately saved via copy running-config startup-config to prevent recurrence
End-device pings failing with no connectivity	IP Configuration left at default (0.0.0.0) on multiple PCs and servers	Static IP addressing applied to every end device per the addressing plan (Section 3.3)
Contractor laptop unable to detect the wireless SSID; fell back to an APIPA address	No wireless network interface installed; IP Configuration set to DHCP with no DHCP service present	Wireless interface module installed; IP Configuration switched to Static with the correct VLAN 40 address

7. Conclusion

This project delivered a fully segmented, centrally routed network for Lesedi Renewable Energy Research Group that satisfies every traced client requirement (R1-R7). The Router-on-a-Stick architecture provides inter-VLAN routing through a single router interface, proportionate to the client's single-site scale; the VLSM addressing plan allocates the assigned block efficiently while preserving capacity for growth; and the restricted contractor wireless VLAN demonstrably enforces the client's explicit requirement for limited, rather than full, after-hours access.

All design decisions were implemented without deviation from the approved Milestone 1 plan, and every claim of functionality in this report is backed by direct, reproducible verification evidence and including the TTL-based proof of inter-VLAN routing and the before/after demonstration of the access control policy. The complete configuration, evidence, and working Packet Tracer project accompany this report in the project's GitHub repository.

8. Recommendations

8.1 Operational Recommendations

- Migrate end-device addressing from static to DHCP for VLANs 10 and 20, reducing administrative overhead as the user population grows.
- Enable port security on access switch ports to mitigate unauthorised device connection.
- Replace open wireless authentication on AP-GUEST with WPA2-PSK or equivalent, as a defence-in-depth measure alongside the existing Layer 3 restriction.
- Migrate device management access from Telnet to SSH, and apply it consistently to the dedicated management VLAN (99).

8.2 Future Architectural Enhancements

The following were explicitly out of scope for this engagement (Section 2.2) but represent natural extensions should the client's needs grow:

- Router and/or core switch redundancy (e.g. HSRP or VRRP), removing the current single points of failure.
- A dynamic routing protocol should the network expand to multiple sites or routers.
- Network Address Translation (NAT/PAT) for a more realistic Internet edge, beyond the point-to-point link used here for verification purposes.
- Formal 802.1X network access control for guest and contractor authentication, if contractor turnover or risk profile increases.

Appendix: Glossary of Terms

Term	Definition
VLAN	Virtual LAN, a logically separate broadcast domain configured on a switch, independent of physical port location.
Trunk port	A switch port carrying multiple VLANs, each frame tagged with its VLAN ID via 802.1Q.
Access port	A switch port carrying exactly one VLAN, untagged, typically connecting to an end device.
802.1Q	The IEEE standard for VLAN tagging on Ethernet frames, enabling a single physical link to carry multiple VLANs.
Router-on-a-Stick	A routing architecture using one physical router interface, divided into logical sub-interfaces, to route between VLANs.
VLSM	Variable Length Subnet Masking, subdividing an address block into differently sized subnets based on actual host requirements.
Wildcard mask	The bitwise inverse of a subnet mask, used in access control lists; a 0 bit must match, a 1 bit is unconstrained.
Extended ACL	An access control list that filters on source and destination address (and optionally protocol/port), applied per interface and direction.
TTL	Time To Live, a packet field decremented at each router hop, usable to infer the number of hops a packet has traversed.